

SYSTEM EXECUTABLES INDEX

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Description of Source Executables

File Name	Location	Description
app.py	Root	Main Flask application server containing routes and logic.
train_model.py	Opticrop - Executable Files & Artifacts	Runs data preprocessing, fits models, and saves pipeline artifacts.
generate_datasets.py	Root	Generates the mock datasets used to train models.
pytest.ini / tests/app.py	Flask application Root	Configuration and scripts for unit testing.

- train_models.py ML training pipeline
- generate_datasets.py Synthetic dataset generator
- tests/test_app.py Automated test suite

2. Output Executables

Model files are saved in the 'models/' directory (e.g. 'best_model.pkl', 'scaler.pkl', 'yield_model.pkl'). Performance charts are saved in 'static/images/'.

- Serialised Models
- models/best_model.pkl RF Classifier pipeline
- models/yield_model.pkl RF Regressor pipeline
- models/scaler.pkl StandardScaler
- models/kmeans_model.pkl K-Means (k=5)
- Configuration
- requirements.txt Pip dependency list
- README.md Setup & run instructions